

Can He

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EDUCATION

Northeastern University - Boston Campus , Boston, MA <i>M.S. in Computer Science, GPA: 3.94 / 4.0</i> Coursework: Computer Systems, Algorithms, Object-Oriented Design, Discrete Structures, Foundations of AI, Machine Learning	Sep 2023 - Apr 2026
Northeastern University - Boston Campus , Boston, MA <i>M.S. in Project Management, GPA: 3.761 / 4.0</i>	Sep 2013 - Jan 2016
Chongqing University of Technology , Chongqing, China <i>B.E. in Mechanical Engineering</i>	Sep 2008 - Jun 2012

TECHNICAL SKILLS

Languages: Python, Java, C/C++, R, MQL4, SQL
AI / Computer Vision: OpenCV, PyTorch, Ultralytics YOLO, Scikit-Learn, NumPy, Pandas, image preprocessing, postprocessing
Backend / Data: RabbitMQ, AWS S3, SQLAlchemy, Alembic, pytest, DAOs, async upload queues, traceability metadata
DevOps / Observability: Docker, Kubernetes, Grafana, Prometheus, systemd, Git, CMake, CI-ready testing workflows

WORK EXPERIENCE

Tristar AI , Boston, MA <i>Junior Software Developer</i> - Own end-to-end production computer vision pipelines for manufacturing customers, converting inspection requirements into high-speed capture, low-latency inference, traceability, PLC feedback, and deployment workflows. - Design and maintain Python services for image preprocessing/postprocessing, task/result schemas, RabbitMQ/AWS S3 publishing, PLC rejection logic, runtime health checks, and deployment configuration. - Harden deployments with pytest coverage, test doubles, monitoring hooks, Grafana/Prometheus metrics, Docker/Kubernetes packaging, and operator runbooks.	May 2026 - Present
Tristar AI , Boston, MA <i>Software Development Intern</i> - Independently implemented data-access objects, upload schedulers, camera/data capture utilities, defect-analysis scripts, configuration loaders, and local test flows for customer-facing AI systems. - Built validation coverage across camera, messaging, storage, and inference boundaries using pytest, dependency injection, local fixtures, and CI-ready project structure.	Jun 2025 - Jan 2026
Northeastern University , Boston, MA <i>Teaching Assistant - DS5220 Supervised Machine Learning</i> Supported ML course delivery by grading assignments, preparing answer keys, explaining supervised-learning concepts, and leading office hours and review sessions.	Sep 2024 - Jun 2025

SELECTED PROJECTS

Amcor Bottle Traceability Pipeline <i>Python, OpenCV, Terpsichore, RabbitMQ, AWS S3, Prometheus, pytest</i> - Owned the full camera-to-cloud traceability pipeline for high-speed bottle production, wiring IPC camera frames, station/team/factory metadata, task lifecycle states, result payloads, and async upload queues. - Implemented low-latency Terpsichore services with Prometheus heartbeat/metrics, RabbitMQ result publishing, AWS S3 evidence uploads, config-driven dev/prod modes, and CI-ready validation. - Built resilient result handling and failure recovery so production evidence, metadata, and monitoring stayed aligned with high-speed line throughput.
Spectrum High-Speed Defect Detection Pipeline <i>Python, OpenCV, PyTorch, Ultralytics YOLO, UNet, RabbitMQ, AWS S3</i> Built the full low-latency inspection pipeline, including IPC camera capture, crop/preprocessing, YOLO/UNet inference wrappers, batch result processing, S3/RabbitMQ outputs, and threshold tuning to drive false-positive rate near 0.01%.
Medegen Manufacturing Defect Detection Pipeline <i>Python, OpenCV, PyTorch, Terpsichore, PLC, RabbitMQ, AWS S3, systemd</i> Owned tub defect detection for four defect categories in a six-camera/three-line production system, maintaining FIFO-triggered capture, low-latency multiprocessing inference, PLC rejection, outbox-based S3/AMQP publishing, and ~0.01% false-positive behavior.
PaddlePaddle RTX 50-Series GPU Compatibility <i>C++, Python, CUDA, CMake, Docker, PaddlePaddle</i> Adapted PaddlePaddle build/dependency configuration for NVIDIA RTX 50-series/Blackwell GPUs, CUDA 12.8/12.9/13.x wheels, architecture flags, and containerized build validation.

EARLIER ENGINEERING EXPERIENCE

Wuhu Automobile Advanced Technology Institute Co., Ltd. , Wuhu, Anhui, China <i>Project Manager</i> Led a 23-person engineering team developing lightweight carbon-fiber automotive components, achieving a 42% weight reduction on targeted Chery Tiggo 8 Pro PHEV components while managing testing, cost control, and production integration.	Sep 2020 - Sep 2023
Chery Holding Group Co., Ltd. , Wuhu, Anhui, China <i>Project Manager</i> Optimized chassis adaptation for fuel and hybrid vehicle lines through testing, model adjustment, technical data management, QC collaboration, patent support, and production guidance.	Sep 2016 - Sep 2020